



ROLL NO

PUNJAB PUBLIC SERVICE COMMISSION
COMBINED COMPETITIVE EXAMINATION
FOR RECRUITMENT TO THE POSTS OF
PROVINCIAL MANAGEMENT SERVICE, ETC -2023
CASE NO. 1C2024

SUBJECT: STATISTICS (PAPER-I)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

NOTE:

Attempt FIVE questions in all. Non-Programmable calculator is allowed.

Q.No.1

(a) The ages of 30 patients admitted to a certain hospital during a particular week are as follows:
 48, 31, 39, 37, 18, 64, 61, 43, 40, 42, 54, 62, 74, 48, 29, 67, 30, 49, 71, 71, 51, 51, 12, 52, 65, 53, 68,
 35, 57, 26, 27, 58

Construct a stem and leaf display from the data and list the data in an array.

(b) Find the missing entries in the following frequency distribution table:

Class Limits	Frequency	Relative Frequency	Cumulative Frequency	Cumulative Percentage
8 To -				25
- To -		0.05		
- To -			9	
- To -		0.30	15	
- To 32				

(12+8=20 Marks)

Q.No.2

(a) A coin and a die are tossed together. Find the probability of getting a head and even number.

(b) If two fair dice are thrown, what is the probability of getting:

(i) Sum between 4 & 8 inclusive.

(ii) Product between 4 & 8 inclusive.

(8+12=20 Marks)

Q.No.3

(a) A man draws 2 balls from a bag containing 3 white and 5 black balls. If he receives Rs.70 for each white ball he draws and Rs.7 for every black ball he draws, find the expectation.

(b) Find the expected value of the r.v x having p.d.f:

$$f(x) = 2(1 - x) \quad 0 < x < 1$$

$$= 0 \quad \text{elsewhere}$$

(10+10=20 Marks)

Q.No.4

(a) A car hire firm has 2 cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distribution with parameter 1.5. Calculate the proportion of days on which some demand is refused. [$e^{-1.5} = 0.2231$].

(b) A random sampling of 4 members of a 150 member club has shown that 3 prefer no smoking in the clubhouse dining room. What is the probability that this will in fact occur if only 20% of the members prefer no smoking in the dining room. Find this probability assuming that the sample was obtained under: (i) Sampling without replacement and (ii) Sampling with replacement. Compare the two.

(10+10=20 Marks)

P.T.O

Q.No.5

(a) The heights of applicants to the police force are normally distributed with mean 170 and standard deviation 3.8cm. If 30% of applicants are rejected because they are too small, what is the minimum accepted height for the police force?

(b) If X is $b(x; 20, 0.4)$ find $P(6 \leq X \leq 10)$. Then find the approximations to this probability using

(i) The Poisson Distribution

(ii) The Normal Distribution.

(10+10=20 Marks)

Q.No.6

Find conditional expectation of bivariate normal distribution.

(20 Marks)

Q.No.7

a) Fit an exponential curve $Y = ae^{bx}$ to the following data:

X	1	2	3	4	5	6
Y	1.6	4.5	13.8	40.2	145.0	363.0

b) Find the normal equations which determine the values of a and b in least squares line $Y = a + bX$; and show that the sum of squares of residuals from the least squares line is given by:

$$\sum Y^2 - a\sum Y - b\sum XY$$

(10+10=20 Marks)

Q.No.8

(a) Given the following data:

X_1	1	4	1	3	2	4
X_2	1	8	3	5	6	10
X_3	2	8	1	7	4	6

Find the least-squares regression line where X_1 is the dependent variable and X_2 and X_3 are independent variables.

b) The following data represent concomitant value~ of three variables:

X_1	32	18	52	16	42	48
X_2	3	2	5	1	4	6
X_3	2	4	2	5	3	9

Calculate all the multiple correlation coefficient, working out the usual simple correlation coefficient.

(10+10=20 Marks)



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SUBJECT: STATISTICS (PAPER-II)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
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NOTE: Attempt any FIVE questions in all. Calculator is allowed. (Not Programmable).

Q.No.1

(a) Given the population: [2,3,4,5,6,7]. Draw all possible samples of size 3 without replacement from the population. Determine the proportion of odd numbers for each sample. Verify that the sample proportion is an unbiased estimate of the population proportion.

(b) Explain in brief the Central limit theorem.

(10+10=20 Marks)

Q.No.2

(a) Explain in brief the process of point estimation by moment methods.

(b) Explain in brief the process of point estimation by maximum likelihood methods.

(10+10=20 Marks)

Q.No.3

A farmer wishes to test the effects of four different fertilizers (A, B, C, and D) on the yield of wheat. In order to eliminate sources of error due to variability in soil fertility, he uses the fertilizers in a Latin-square arrangement, as shown in the following table where the numbers indicate yields in bushels per unit area. Perform an analysis of variance to determine whether there is a difference between the fertilizers at significance level of 0.05.

A 9	C 12	D 16	B 2
D 13	B 3	A 6	C 10
B 6	A 11	C 14	D 15
C 13	D 12	B 1	A 8

(20 Marks)

Q.No.4

(a) Explain in brief the components of a time series.

(b) The following table shows the monthly new housing starts (in thousands) for a certain country from January 1990 through December 1992. Obtain a seasonal index by using the average-percentage method.

Month	1990	1991	1992
January	99.2	52.5	71.6
February	86.9	59.1	78.8
March	108.5	73.8	111.6
April	119.0	99.7	107.6
May	121.1	97.7	115.2
June	117.8	103.4	117.8
July	111.2	103.5	106.2
August	102.8	94.7	109.9
September	93.1	86.6	106.0
October	94.2	101.8	111.8
November	81.4	75.6	84.5
December	57.4	65.6	78.6

(10+10=20 Marks)

P.T.O

Q.No.5 (a) Explain the concept of an index number. Also discuss the main steps involved in the construction of index numbers of wholesale prices.

(b) Given the data below:

Year	Commodity Price (in Rs.)			
	A	B	C	D
2019	16.25	20.00	24.00	10.50
2020	17.50	22.50	27.00	12.00
2021	21.00	29.00	32.50	15.00
2022	29.00	36.00	35.50	18.00

Compute the index numbers by taking the (i) Mean (ii) Geometric mean of price relatives.
Consider 2019 as base year. **(6+14=20 Marks)**

Q.No.6 Enlist the statistical organizations in Pakistan and discuss their roles. **(20 Marks)**

Q.No.7 (a) Describe the system for collection of vital statistics in Pakistan.

(b) Calculate the Crude and Standardized death rates of 2022 population by (i) direct and (ii) indirect methods, using 2000 population as a standard.

Age group (Years)	2000		2022	
	Population	Deaths	Population	Deaths
0 – 9	10000	220	15000	20
10 – 19	11000	132	12000	11
20 – 49	7000	105	8000	14
50 & over	2000	90	5000	42

(5+15=20 Marks)

Q.No.8 (a) Define Analysis of Variance and its assumptions.

(b) Given the following observations recorded from three groups

A:	47.2	49.8	48.5	48.7	
B:	50.1	49.3	51.5	50.9	
C:	49.1	53.2	51.2	52.8	52.3

Test the hypothesis that means of three groups are equal. If appropriate, apply least significant test to identify the significantly different group(s). Use 5% level of significance. **(6+14=20 Marks)**